

# Project WalkThrough

## CodeBase

## ML Basics

## Report

### 1. The Core Problem (Introduction)

Amyotrophic Lateral Sclerosis (ALS) is a severe disease that breaks down motor neurons in the nervous system. Right now, doctors struggle to diagnose it early because there is no simple, non-invasive blood test.

Scientists can look at blood using "transcriptomics" (studying which genes are turned on or off), but the genetic signals for ALS in the blood are incredibly weak and scattered.

Because the signals are so quiet, a single study or a single dataset usually doesn't have enough statistical power to find them reliably. To fix this, your project aimed to combine massive amounts of data from different sources to create one powerful, universal "molecular fingerprint" for ALS.

### 2. The Data (Materials)

The project used two different datasets of peripheral whole blood, representing two different generations of genetic technology:

- Microarray (Older Tech): Think of this like taking a photograph of the genes using chemical tags that glow. The dataset (GSE112681) had 1,115 samples.
- RNA-Sequencing (Newer Tech): Think of this like literally counting every single genetic letter. The dataset (GSE234297) had 144 samples.

### 3. The Pipeline (Pre-processing & Harmonization)

Combining these two datasets is like trying to compare a digital photograph with a hand-drawn painting; they measure the same thing but look completely different mathematically.

Here is how the project solved that:

- Finding Common Ground: The team matched the genes between the two datasets, finding an overlapping space of 9,344 shared genes.
- Batch-Effect Correction (ComBat): This is the most crucial step. When you merge data from two different machines, the mathematical differences caused by the machines (technical noise) can drown out the actual biological differences (the disease). The team used an algorithm called ComBat to digitally erase the machine-specific biases while preserving the true ALS signals.

#### 4. Biological Analysis (Finding the Genes)

Once the data was combined and cleaned, the team used statistical math (a tool called [limma](#)) to find which genes were behaving differently in ALS patients versus healthy people.

- The Fingerprint: They found a strict "core" of 4 highly significant genes, a medium-confidence group of about 90 genes, and a broader set of 500 genes to use for machine learning.
- Network Analysis (WGCNA): The team tried to see if these genes grouped together into functional "neighborhoods" or modules. However, because the ALS signal in blood is so subtle, this network analysis only found one weak group and wasn't very useful.

#### 5. Machine Learning (The Ultimate Test)

To prove this 500-gene fingerprint actually works, the team set up a brutal, cross-platform test. They trained Artificial Intelligence models using *only* the RNA-seq data, and then tested them on the completely unseen Microarray data. This ensures the AI is learning the actual disease, not just memorizing the data.

They tested three AI models:

1. Random Forest: A complex "council" of decision trees. It scored an AUC of 0.669 (AUC is a grade from 0.5 to 1.0, where 1.0 is perfect). It struggled slightly because it over-complicated things and memorized some technical noise.
2. Support Vector Machine (SVM): A model that tries to draw a mathematical boundary between healthy and sick patients. It did slightly better with an AUC of 0.682.
3. Logistic Regression: A simpler, linear mathematical scale. Surprisingly, this simple model performed the best, scoring an AUC of 0.712!. It proved that for subtle genetic signals, a simple, straightforward model is less likely to get confused by noise.

#### 6. The Ablation Study (Finding the Sweet Spot)

Finally, the team did an "Ablation Study" on the winning Logistic Regression model. This means they slowly stripped away genes to see how many were *actually* needed.

They found a "Goldilocks zone": exactly 25 genes provided the highest accuracy (an AUC of 0.746). Using 50 or 100 genes actually made the model worse because it started picking up unnecessary technical noise.

## Summary

In short, this project successfully proved that by carefully cleaning and combining different types of blood data, we can use a simple machine learning model to find a reliable 25-gene signature that spots ALS, paving the way for easier diagnoses.

## Abstract

Amyotrophic Lateral Sclerosis (ALS) is hard to diagnose early because the disease's genetic signs in the blood are extremely faint and scattered. To fix this, the project mathematically combined two massive, different sets of genetic data to filter out technical errors and make those hidden biological signals louder. By training machine learning (AI) on this cleaned-up data, the study successfully found a "molecular fingerprint" of genes that can help identify the disease. This proves that combining different types of data can pave the way for a simple, non-invasive blood test to diagnose ALS

## Introduction

Amyotrophic lateral sclerosis (ALS) is a devastating nervous system disease, and early clinical intervention is hindered by the lack of reliable, non-invasive blood biomarkers. While high-throughput transcriptomics offers a way to profile molecules, the biological indicators of ALS in peripheral blood are extremely faint and widely distributed. Consequently, single-platform studies—using only microarrays or only RNA-sequencing—lack the statistical power to reliably detect these elusive signals. Although merging data from multiple platforms can improve detection, this process inherently introduces a dramatic amount of non-biological technical variation. Merging datasets without rigorous algorithmic batch-effect correction drastically inflates false-positive rates and obscures true biological signals. To overcome these challenges, this study introduces a multidimensional computational pipeline that seamlessly integrates and batch-corrects large-scale RNA-seq and microarray transcriptomes. This rigorous algorithmic approach resolves both the sensitivity limits of single-cohort studies and the technical hurdles of uncorrected data merging. Ultimately, this methodology yields a highly reliable, all-encompassing molecular signature that can be used universally for the non-invasive diagnosis of ALS.